3.1.1 Choosing the right basis

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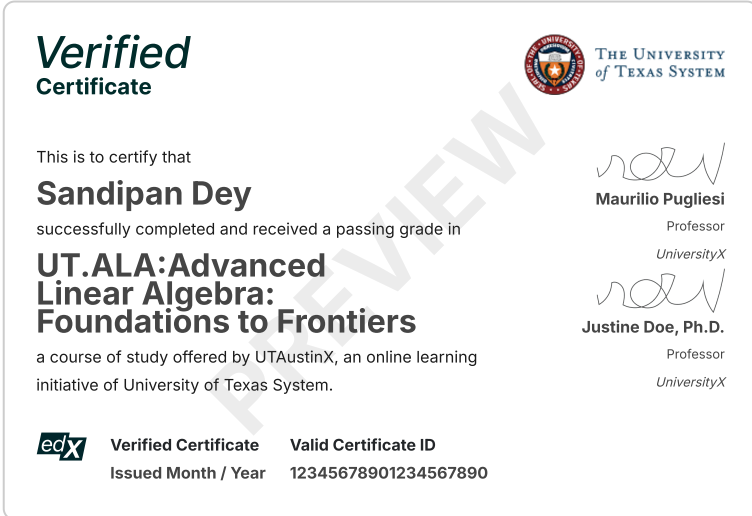

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equations, we could

instead write this as the following matrix.

Let's see these are the columns of the matrix. The first column is all ones because

it's really chi 0 to the zeroth power, chi 1 to the zeroth power, and so forth. So

here we have the matrix which must multiply gamma 0, gamma 1, and so forth

through gamma n minus 1. And that should be equal to phi 0, phi 1, through phi n-1.

Okay, but again

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Reading assignment

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Read Unit 3.1.1 of the notes. [LINK]

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? HW 3.1.1.1. Shouldn't condition number of X be 1 for n = 0?
Shouldn't [mathjaxinline] \kappa_2(X)=1[/mathjaxinline] when [mathjaxinline... 6

Homework 3.1.1.1

1/1 point (graded)

Choose $\chi_0, \chi_1, \cdots, \chi_{m-1}$ to be equally spaced in the interval $[0,1]$: for $i = 0, \ldots, m-1$, $\chi_i = ih$,where $h = 1/(m-1)$. Write Matlab code to create the matrix

$$X = \begin{pmatrix} 1 & \chi_0 & \cdots & \chi_0^{n-1} \\ 1 & \chi_1 & \cdots & \chi_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \chi_{m-1} & \cdots & \chi_{m-1}^{n-1} \end{pmatrix}$$

as a function of n with $m = 5000$. Plot the condition number of X, $\kappa_2(X)$, as a function of n (Matlab's function for computing $\kappa_2(X)$ is cond(X) .)

☑ done

✔

For full solution see notes: [LINK]

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Answers are displayed within the problem

Homework 3.1.1.2

1/1 point (graded)

Choose $\chi_0, \chi_1, \cdots, \chi_{m-1}$ to be equally spaced in the interval $[0,1]$: for $i = 0, \ldots, m-1$, $\chi_i = ih$, where $h = 1/(m-1)$. Write Matlab code to create the matrix

$$X = \begin{pmatrix} 1 & P_1(\chi_0) & \cdots & P_{n-1}(\chi_0) \\ 1 & P_1(\chi_1) & \cdots & P_{n-1}(\chi_1) \\ \vdots & \vdots & & \vdots \\ 1 & P(\chi_{m-1}) & \cdots & P_{n-1}(\chi_{m-1}) \end{pmatrix}$$

as a function of n with $m = 5000$. Plot $\kappa_2(X)$ as a function of n. To check whether the columns of X are mutually orthogonal, report $\|X^T X - D\|_2$ where D equals the diagonal of $X^T X$.

☑ done

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of kind of looks like an inner product doesn't it? And it kind of looks like a

orthogonality condition. So what you can then do is do the exact same

thing that we did

before, which is create a matrix with as its columns the

polynomial of degree 0 evaluated at chi 0, chi 1 and so forth and then the

polynomial with degree 1 evaluated at chi 0, chi,

and so forth. And then multiplying that times the vector that now become the

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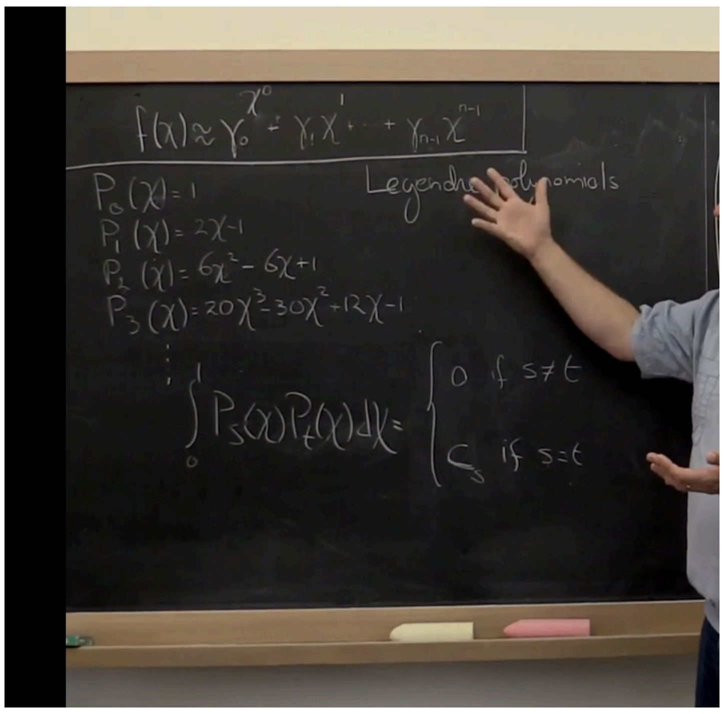

What we've just seen is an example where if you formulate the problem better, in

this case, if you make it so that the family of polynomials with which you

create your n minus first degree polynomial, if that

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